

Translated

From: e-maxx.ru

Delaunay triangulation and Voronoi diagram

Consider a set $\{p_i\}$ of points on the plane. A **Voronoi diagram** $V(\{p_i\})$ of $\{p_i\}$ is a partition of the plane into n regions V_i , where $V_i = \{p \in \mathbb{R}^2; \rho(p, p_i) = \min \rho(p, p_k)\}$. The cells of the Voronoi diagram are polygons (possibly infinite). A **Delaunay triangulation** $D(\{p_i\})$ of $\{p_i\}$ is a triangulation where every point p_i is outside or on the boundary of the circumcircle of each triangle $T \in D(\{p_i\})$.

There is a nasty degenerated case when the Voronoi diagram isn't connected and Delaunay triangulation doesn't exist. This case is when all points are collinear.

Properties

The Delaunay triangulation maximizes the minimum angle among all possible triangulations.

The Minimum Euclidean spanning tree of a point set is a subset of edges of its' Delaunay triangulation.

Duality

Suppose that $\{p_i\}$ is not collinear and among $\{p_i\}$ no four points lie on one circle. Then $V(\{p_i\})$ and $D(\{p_i\})$ are dual, so if we obtain one of them, we may obtain the other in $O(n)$. What to do if it's not the case? The collinear case may be processed easily. Otherwise, V and D' are dual, where D' is obtained from D by removing all the edges such that two triangles on this edge share the circumcircle.

Building Delaunay and Voronoi

Because of the duality, we only need a fast algorithm to compute only one of V and D . We will describe how to build $D(\{p_i\})$ in $O(n \log n)$. The triangulation will be built via divide-and-conquer algorithm due to Guibas and Stolfi.